Credit Name: CSE2130 - File Structures & Exception Handling

Assignment Name: CountVowels

Error Log Entry

What error message did you encounter (if any)?

```
Exception in thread "main" java.lang.StringIndexOutOfBoundsException: Index 5 out of bounds for length 5
        at java.base/jdk.internal.util.Preconditions$1.apply(Preconditions.java:55)
        at java.base/jdk.internal.util.Preconditions$1.apply(Preconditions.java:52)
        at java.base/jdk.internal.util.Preconditions$4.apply(Preconditions.java:213)
        at java.base/jdk.internal.util.Preconditions$4.apply(Preconditions.java:210)
        at java.base/jdk.internal.util.Preconditions.outOfBounds(Preconditions.java:98)
        at java.base/jdk.internal.util.Preconditions.outOfBoundsCheckIndex(Preconditions.java:106)
        at java.base/jdk.internal.util.Preconditions.checkIndex(Preconditions.java:302)
        at java.base/java.lang.String.checkIndex(String.java:4832)
        at java.base/java.lang.StringLatin1.charAt(StringLatin1.java:46)
        at java.base/java.lang.String.charAt(String.java:1555)
        at Mastery.countVowels.main(countVowels.java:42)
```

What unexpected behavior did your program exhibit?

1. Error Message occured because the amount of vowels(letters) was shorter than the text

What caused the issue? (e.g., syntax error, logic error, incorrect function usage, etc.)

1.Logic Error - Having letter.charAt() with i as well makes it so that it doesn't compare all the letters in the lowercase text with every single vowel, but rather only one vowel. Which may also not work if the word does not have the same amount of letters as the vowels (5)

Include a screenshot of specific lines of code.

```
if(lowercaseText.charAt(i) == letter.charAt(i))
{
```

How did you fix the issue?

1. I changed the 'i' to the specific vowels so that it can compare every letter to every single vowel.

Provide the corrected code or solution using a screenshot.

```
if(lowercaseText.charAt(i) == letter.charAt(0))
{
    //update total vowels
    vowelSum += 1;
    a += 1;
}
if(lowercaseText.charAt(i) == letter.charAt(1))
{
    //update total vowels
    vowelSum += 1;
    E += 1;
}
if(lowercaseText.charAt(i) == letter.charAt(2))
{
    //update total vowels
    vowelSum += 1;
    I += 1;
}
if(lowercaseText.charAt(i) == letter.charAt(3))
{
    //update total vowels
    vowelSum += 1;
    o += 1;
}
if(lowercaseText.charAt(i) == letter.charAt(4))
{
    //update total vowels
    vowelSum += 1;
    u += 1;
}
```